

```
1 name: Matrix Demo
2 on: push
3 jobs:
4   build:
5     continue-on-error: true
6     strategy:
7       matrix:
8         node-version: [12, 14, 16]
9         operating-system: [ubuntu-latest, windows-latest]
10      include:
11        - node-version: 18
12          operating-system: ubuntu-latest
13      exclude:
14        - node-version: 12
15          operating-system: windows-latest
16      runs-on: ${{ matrix.operating-system }}
17      steps:
18        - name: Get Code
19          uses: actions/checkout@v3
20        - name: Install NodeJS
21          uses: actions/setup-node@v3
22          with:
23            node-version: ${{ matrix.node-version }}
24        - name: Install Dependencies
25          run: npm ci
```

need full code

Based on the screenshot, here's the complete `matrix.yml` workflow.

```
name: Matrix Demo

on: push

jobs:
  build:
    continue-on-error: true

    strategy:
      matrix:
        node-version: [12, 14, 16]
        operating-system: [ubuntu-latest, windows-latest]

        include:
          - node-version: 18
            operating-system: ubuntu-latest

        exclude:
          - node-version: 12
            operating-system: windows-latest

    runs-on: ${{ matrix.operating-system }}

    steps:
      - name: Get Code
        uses: actions/checkout@v3
      - name: Install NodeJS
        uses: actions/setup-node@v3
        with:
          node-version: ${{ matrix.node-version }}
      - name: Install Dependencies
        run: npm ci
```

</> YAML



```
uses: actions/checkout@v3

- name: Install NodeJS
  uses: actions/setup-node@v3
  with:
    node-version: ${{ matrix.node-version }}

- name: Install Dependencies
  run: npm ci

- name: Run Tests
  run: npm test

- name: Build Project
  run: npm run build

- name: Output Information
  run: |
    echo "Operating System: ${{ matrix.operating-system }}"
    echo "Node Version: ${{ matrix.node-version }}"
```

Matrix combinations generated

The matrix will execute these jobs:

Node Version	Operating System	Status
12	ubuntu-latest	✓ Runs
12	windows-latest	✗ Excluded
14	ubuntu-latest	✓ Runs
14	windows-latest	✓ Runs
16	ubuntu-latest	✓ Runs
16	windows-latest	✓ Runs
18	ubuntu-latest	✓ Added via <code>include</code>

Total jobs executed: 6

- `include` adds an extra matrix combination (Node 18 on Ubuntu).
- `exclude` removes the `Node 12 + Windows` combination.
- `continue-on-error: true` allows the workflow to continue even if one of the matrix jobs fails. GitHub evaluates `include` and `exclude` as part of matrix expansion.

pls explain the steps

Let's go through this workflow **line by line**. Matrix strategies are one of the most powerful features of GitHub Actions because they let you run the same job with different configurations.

Complete Workflow

</> YAML



```
name: Matrix Demo

on: push

jobs:
  build:
    continue-on-error: true

    strategy:
      matrix:
        node-version: [12, 14, 16]
        operating-system: [ubuntu-latest, windows-latest]

        include:
          - node-version: 18
            operating-system: ubuntu-latest

        exclude:
          - node-version: 12
            operating-system: windows-latest

    runs-on: ${ matrix.operating-system }

    steps:
      - name: Get Code
        uses: actions/checkout@v3

      - name: Install NodeJS
```

```
uses: actions/setup-node@v3
with:
  node-version: ${{ matrix.node-version }}

- name: Install Dependencies
  run: npm ci

- name: Run Tests
  run: npm test

- name: Build Project
  run: npm run build

- name: Output Information
  run: |
    echo "Operating System: ${{ matrix.operating-system }}"
    echo "Node Version: ${{ matrix.node-version }}"
```

Step 1: Workflow Name

</> YAML



```
name: Matrix Demo
```

This is simply the name displayed in GitHub Actions.

GitHub UI

Actions



Matrix Demo

Step 2: Trigger

</> YAML



```
on: push
```

Whenever you push code,

```
git push origin main
```



this workflow starts automatically.

Step 3: Job

</> YAML



```
jobs:
  build:
```

Creates one job named **build**.

Without a matrix, only **one** build job runs.

With a matrix, GitHub creates **multiple build jobs automatically**.

Step 4: Continue on Error

</> YAML



```
continue-on-error: true
```

Suppose one matrix job fails.

Normally

Node 12 ❌



the entire workflow becomes failed.

With

</> YAML



```
continue-on-error: true
```

GitHub keeps executing the remaining matrix jobs.

Example

Ubuntu + Node12 ❌



Ubuntu + Node14 ✅

Ubuntu + Node16 

Windows + Node14 

Windows + Node16 

Workflow continues.

Step 5: Matrix Strategy

 YAML 

```
strategy:
```

This tells GitHub

"Run this same job multiple times."

Step 6: Matrix Variables

 YAML 

```
matrix:
```

Here we define variables.

First Variable

 YAML 

```
node-version:  
  - 12  
  - 14  
  - 16
```

Means

Run using Node 12 

Run using Node 14

Run using Node 16

Second Variable

YAML

```
operating-system:  
  - ubuntu-latest  
  - windows-latest
```

Means

Run on Ubuntu

Run on Windows

Step 7: Cartesian Product

GitHub combines every value with every other value.

Think of multiplication.

Node Versions

12
14
16

×

Operating Systems

Ubuntu
Windows

Result

Node

OS

12

Ubuntu

Node	OS
12	Windows
14	Ubuntu
14	Windows
16	Ubuntu
16	Windows

Total

3 × 2 = 6 jobs



GitHub automatically creates all six.

Step 8: Include

</> YAML



```
include:  
  - node-version: 18  
    operating-system: ubuntu-latest
```

This adds an extra job.

Normally

```
12 Ubuntu  
12 Windows  
  
14 Ubuntu  
14 Windows  
  
16 Ubuntu  
16 Windows
```



After include

```
18 Ubuntu
```



gets added.

Now

7 jobs



exist.

Step 9: Exclude

</> YAML



```
exclude:  
  - node-version: 12  
    operating-system: windows-latest
```

Removes one combination.

Deletes

Node12 + Windows



Now jobs become

12 Ubuntu



14 Ubuntu

14 Windows

16 Ubuntu

16 Windows

18 Ubuntu

Total

6 jobs



Step 10: runs-on

</> YAML



```
runs-on: ${{ matrix.operating-system }}
```

Instead of writing

</> YAML



```
runs-on: ubuntu-latest
```

we use

</> YAML



```
matrix.operating-system
```

GitHub replaces it.

Job 1

</> YAML



```
runs-on: ubuntu-latest
```

Job 2

</> YAML



```
runs-on: windows-latest
```

Job 3

</> YAML



```
runs-on: ubuntu-latest
```

and so on.

Step 11: Checkout

</> YAML



```
uses: actions/checkout@v3
```

Downloads your repository.

Without this

No code



exists on the runner.

Step 12: Setup Node

</> YAML



```
uses: actions/setup-node@v3
```

Installs Node.js.

Then

</> YAML



```
with:  
  node-version: ${{ matrix.node-version }}
```

changes according to the matrix.

Job 1

Install Node 12



Job 2

Install Node 14



Job 3

Install Node 16



Job 4

Install Node 18



Step 13: Install Dependencies

</> YAML



```
run: npm ci
```

Runs

```
npm ci
```



which installs packages from

```
package-lock.json
```



Step 14: Run Tests

</> YAML



```
run: npm test
```

Executes your test suite.

Example

```
Jest
```



```
Mocha
```

```
Vitest
```

Step 15: Build

</> YAML



```
run: npm run build
```

Runs the build script.

For example

Angular



React

Node

Next.js

Step 16: Output Information

</> YAML



```
run: |
```

Runs multiple shell commands.

</> YAML



```
echo "Operating System: ${ matrix.operating-system }"
```

prints

```
Operating System: ubuntu-latest
```



or

```
Operating System: windows-latest
```



Then

</> YAML



```
echo "Node Version: ${ matrix.node-version }"
```

prints

```
Node Version: 14
```



etc.

Final Execution Flow



Push



Matrix Created



12 + Ubuntu



Checkout

Install Node 12

npm ci

npm test

npm run build

14 + Ubuntu



Checkout

Install Node 14

npm ci

npm test

npm run build

14 + Windows



Checkout

Install Node 14

npm ci

npm test

npm run build

16 + Ubuntu

```
graph TD; A[ ] --> B[Checkout<br/>Install Node 16<br/>npm ci<br/>npm test<br/>npm run build]; B --> C[16 + Windows]; C --> D[Checkout<br/>Install Node 16<br/>npm ci<br/>npm test<br/>npm run build]; D --> E[18 + Ubuntu]; E --> F[Checkout<br/>Install Node 18<br/>npm ci<br/>npm test<br/>npm run build];
```

Checkout
Install Node 16
npm ci
npm test
npm run build

16 + Windows

Checkout
Install Node 16
npm ci
npm test
npm run build

18 + Ubuntu

Checkout
Install Node 18
npm ci
npm test
npm run build

Summary

- `strategy.matrix` creates multiple job configurations automatically.
- `node-version` and `operating-system` are matrix variables.
- GitHub generates all combinations (Cartesian product) of the matrix values.
- `include` adds extra combinations that aren't part of the default product.
- `exclude` removes unwanted combinations.
- `runs-on: ${matrix.operating-system}}` and `node-version: ${matrix.node-version}` substitute the values for each generated job.
- Each matrix job runs independently and, by default, in parallel, which makes it ideal for testing across multiple environments.



Do you like this personality?

